

An analysis of von Economo neurons in the cerebral cortex of cetaceans, artiodactyls, and perissodactyls

Mary Ann Raghanti · Linda B. Spurlock · F. Robert Treichler ·
Sara E. Weigel · Raphaela Stimmelmayer · Camilla Butti ·
J. G. M. Hans Thewissen · Patrick R. Hof

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Abstract Von Economo neurons (VENs) are specialized projection neurons with a characteristic spindle-shaped soma and thick basal and apical dendrites. VENs have been described in restricted cortical regions, with their most frequent appearance in layers III and V of the anterior cingulate cortex, anterior insula, and frontopolar cortex of humans, great apes, macaque monkeys, elephants, and some cetaceans. Recently, a ubiquitous distribution of VENs was reported in various cortical areas in the pygmy hippopotamus, one of the closest living relatives of cetaceans. That finding suggested that VENs might not be unique to only a few species that possess enlarged brains.

In the present analysis, we assessed the phylogenetic distribution of VENs within species representative of the superordinal clade that includes cetartiodactyls and perissodactyls, as well as afrotherians. In addition, the distribution of fork cells that are often found in close proximity to VENs was also assessed. Nissl-stained sections from the frontal pole, anterior cingulate cortex, anterior insula, and occipital pole of bowhead whale, cow, sheep, deer, horse, pig, rock hyrax, and human were examined using stereologic methods to quantify VENs and fork cells within layer V of all four cortical regions. VENs and fork cells were found in each of the species examined here with species-specific differences in distributions and densities. The present results demonstrated that VENs and fork cells were not restricted to highly encephalized or socially complex species, and their repeated emergence among distantly related species seems to represent convergent evolution of specialized pyramidal neurons. The widespread phylogenetic presence of VENs and fork cells indicates that these neuron morphologies readily emerged in response to selective forces, whose variety and nature are yet to be identified.

M. A. Raghanti (✉) · L. B. Spurlock · S. E. Weigel
Department of Anthropology and School of Biomedical
Sciences, Kent State University, 750 Hilltop Drive,
222 Lowry Hall, Kent, OH 44242, USA
e-mail: mraghant@kent.edu

F. Robert Treichler
Department of Psychology, Kent State University, Kent,
OH, USA

R. Stimmelmayer
Department of Wildlife Management, North Slope Borough,
Barrow, AK, USA

C. Butti · P. R. Hof
Fishberg Department of Neuroscience and Friedman Brain
Institute, Icahn School of Medicine at Mount Sinai,
New York, NY, USA

J. G. M. Hans Thewissen
Department of Anatomy and Neurobiology, Northeast Ohio
Medical University, Rootstown, OH, USA

P. R. Hof
New York Consortium in Evolutionary Primatology,
New York, NY, USA

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Introduction

The von Economo neuron (VEN) is a specialized projection neuron characterized by a spindle-shaped soma and thick basal and apical dendrites. These cells were first described in the anterior cingulate and insular cortex of humans and were initially referred to as spindle cells (Betz

1881; von Economo and Koskinas 1925; von Economo 1926). They were subsequently re-named in honor of Constantin von Economo (Allman et al. 2005) and have generated significant interest and concern over the past two decades as a potential neuroanatomical substrate that might contribute to uniquely human cognitive processing (e.g., Allman et al. 2005).

Nimchinsky and colleagues redescribed and quantified the distribution of VENs within the human anterior cingulate cortex (ACC) (Nimchinsky et al. 1995), and subsequently reported their presence in great apes (chimpanzees, bonobos, gorillas, and orangutans) and apparent absence in monkeys (Nimchinsky et al. 1999). In both humans and great apes, VENs are localized within layers III and V of restricted cortical regions, including the ACC and insular cortex. The frontoinsular cortex (FI, the most anterior ventral agranular portion of the insula) is a subdomain described in humans and great apes that is defined by characteristic clusters of VENs in layer Vb (Allman et al. 2010; von Economo 1926). Initially, it was posited that VENs evolved from an ancient neuron type specifically within this phylogenetic group to support cognitive or behavioral specializations (Nimchinsky et al. 1999; Allman et al. 2010; Seeley et al. 2012). Their almost exclusive presence within the FI and ACC suggested a significant role in social awareness and interoception (Craig 2009; Allman et al. 2005, 2010). Further studies reported significant decreases in the number of VENs associated with the behavioral variant of frontotemporal dementia (Seeley et al. 2006; Kim et al. 2012; Seeley et al. 2007a; Seeley 2008). Similarly, decreases in VENs were found to be related to agenesis of the corpus callosum (Kaufman et al. 2008), and also to schizophrenia (Brüne et al. 2010, 2011). Both increased and decreased numbers of VENs have also been associated with autism spectrum disorders, although the varied phenotypes associated with this disorder have made interpretation of these heterogeneous results difficult (Kennedy et al. 2007; Simms et al. 2009; Allman et al. 2005; Santos et al. 2011). Interestingly, elderly humans who did not exhibit a typical age-related cognitive decline possessed a fourfold higher density of VENs in the ACC relative to cognitively normal elderly individuals (Rogalski et al. 2013). Taken together, these lines of evidence suggested a potentially contributory role for VENs to cognitive processing in humans.

After VENs were reported in humans and great apes, these neurons were also discovered in some cetaceans. However, the cortical distribution noted in cetacean species sometimes included the ACC, insular cortex, frontal pole, inferior temporal cortex, occipital region of the ecto- and suprasylvian cortex, and the paralimbic cortex (Hof and Van der Gucht 2007; Butti et al. 2009). Shortly thereafter, VENs were also observed in both the African and Indian

elephants (Hakeem et al. 2009). As in other species, distribution was anatomically limited, with VENs present in the anterior insula (AI), frontal pole, and dorsolateral frontal area of the African elephant, and in the ACC of the Indian elephant (Hakeem et al. 2009).

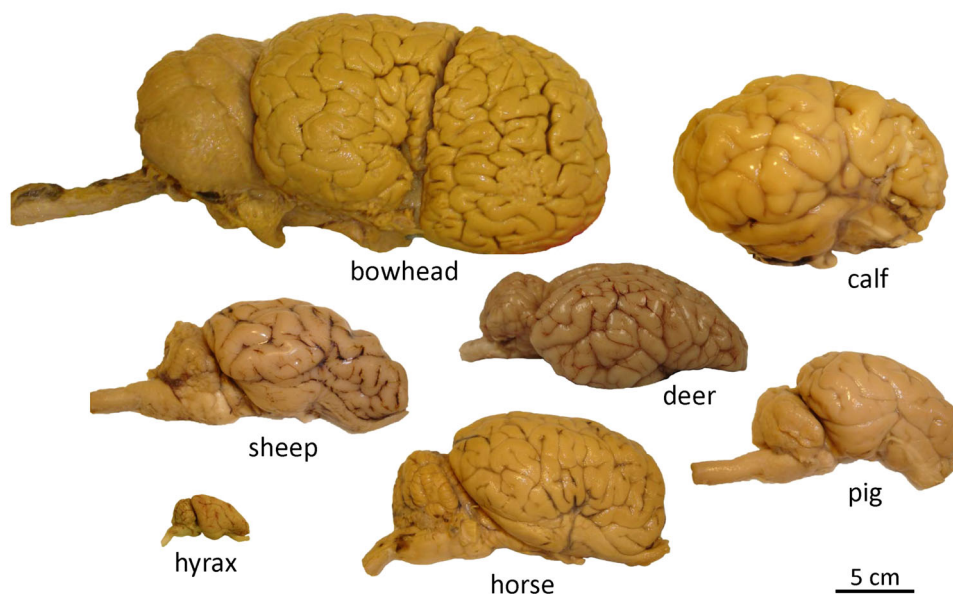
The discovery of VENs in elephants and cetaceans, along with their presence in humans and great apes, led to hypotheses that these neurons were characteristic of large brains and possibly, of socially complex animals. The presence of VENs in these distantly related species suggested the prospect of convergent evolution, in which VENs independently arose to facilitate rapid information processing in enlarged brains. However, more recent findings have challenged these interpretations. VENs have been found within the anterior insular cortex of macaque monkeys (Evrard et al. 2012; Bauernfeind et al. 2013) and Butti et al. reported a ubiquitous distribution of VENs in the neocortex of the pygmy hippopotamus, one of the closest living relatives of cetaceans (Butti et al. 2014, 2011).

With the findings of VENs in the pygmy hippopotamus and macaque, it is possible that these cells represent a phylogenetically ancient neuron type rather than an emergent specialized neuron. An additional distinctive layer V neuron type that was commonly observed in proximity to VENs in the human frontoinsular cortex is the fork cell. Fork cells (Gabelzellen; Ngowyang 1932) are characterized by a single large basal dendrite and two large ‘forked’ apical dendrites. These neurons are thought to have a restricted phylogenetic distribution that is closely linked to the presence of VENs (reviewed in Seeley et al. 2012), but fork cells have not been previously examined in a wide range of species. One objective of the present analysis was to assess the number and spatial distribution of VENs and fork cells in additional species representative of the superordinal clade that includes cetartiodactyls and perissodactyls, as well as afrotherians.

Methods

The sample included representative species of the cetartiodactyls: bowhead whale (*Balaena mysticetus*), domesticated pig (*Sus scrofa domesticus*), sheep (*Ovis aries*), cow (calf; *Bos taurus*), white-tailed deer (*Odocoileus virginianus*); afrotherians: rock hyrax (*Procavia capensis*); perissodactyls: horse (*Equus ferus caballus*); and the human (*Homo sapiens*) (Fig. 1). The pig, sheep, cow, and horse brains were purchased from a commercial vendor (Ward’s Science, Rochester, NY), the rock hyrax brain was obtained from the Cleveland Metroparks Zoo (Cleveland, Ohio), the white-tailed deer brain was acquired as part of

Fig. 1 Macroscopic views of the right hemisphere for each of the species examined. The cerebellum and brainstem were removed from the calf brain and the brain of the bowhead whale was cut into two blocks prior to our study. Scale bar 5 cm



an official deer population cull in Northeast Ohio (permit #6739695382), the bowhead whale brain was obtained from the Department of Wildlife Management, North Slope Borough, Barrow, AK, and human brain specimens were from the Northwestern University Alzheimer's Disease Center Brain Bank. These brains were all from adults, with the exception of the cow. Only one individual per species was analyzed in this study.

All postmortem brains were immersion-fixed in 10 % buffered formalin and the samples for this study were from the right hemisphere. Prior to sectioning, specimens were cryoprotected in a series of sucrose solutions (10, 20, and 30 %), frozen on dry ice, and cut into 40 or 60 μm -thick sections using a sliding microtome (Leica SM2000R, Buffalo Grove, IL). Every 10th section was stained for Nissl substance using 0.05 % cresyl violet solution. The four cortical regions examined included the frontal pole, ACC, AI, and occipital pole (Fig. 2). These regions were chosen to provide consistency in anatomical sampling among diverse species and no homology of function was assumed.

Quantitative analyses were performed using computer-assisted stereology with a system including an Olympus BX-51 photomicroscope equipped with a digital camera and Stereo Investigator software (MBF Biosciences, Williston, VT, version 11). The presence of VENs and fork cells was qualitatively noted throughout all layers of each cortical region. We chose to quantify the density of VENs and fork cells within layer V because this is the cortical layer where these neurons have most frequently been reported. The identification of VENs was based on their elongated, spindle-shaped somata with thick basal and apical dendrites. Spindle-shaped, or fusiform, neurons that

were not appreciably larger than surrounding pyramidal neurons were not categorized as VENs. Fork cell identification was based on their unique morphological appearance that includes a distinct basal dendrite and bifid apical dendrites. The densities of VENs, fork cells, and total neurons within layer V were obtained using the optical fractionator. Total neuron population estimates were included to facilitate comparisons among species and to calculate the percentage of VENs and fork cells. Layer V was outlined for each cortical region at low magnification (4x, N.A. 0.13) and quantified at high magnification (40x, N.A. 0.75). Counting frames were set at $100 \times 100 \mu\text{m}$ and neurons were counted when the nucleolus was in focus within the counting frame. The counting frame height was set at 7 μm to allow for 2- μm guard zones at the top and bottom of sections, and section thickness was measured at every fifth sampling site. Sampling grids were set for each species to obtain approximately 20 sampling sites per section, with a minimum of three sections per area per individual, and to maintain a coefficient of error below 0.08 for each data point.

Results

VENs and fork cells were found in each of the species examined here, and in agreement with previous reports from other species (Nimchinsky et al. 1999; Hakeem et al. 2009; Evrard et al. 2012; Butti et al. 2009, 2011, 2014), were most commonly observed in layers III and V (Fig. 3). We also found VENs in layer II of several of these species (bowhead whale, sheep, calf, pig, and deer), with clusters of layer II VENs present in the sheep and cow (Fig. 4).

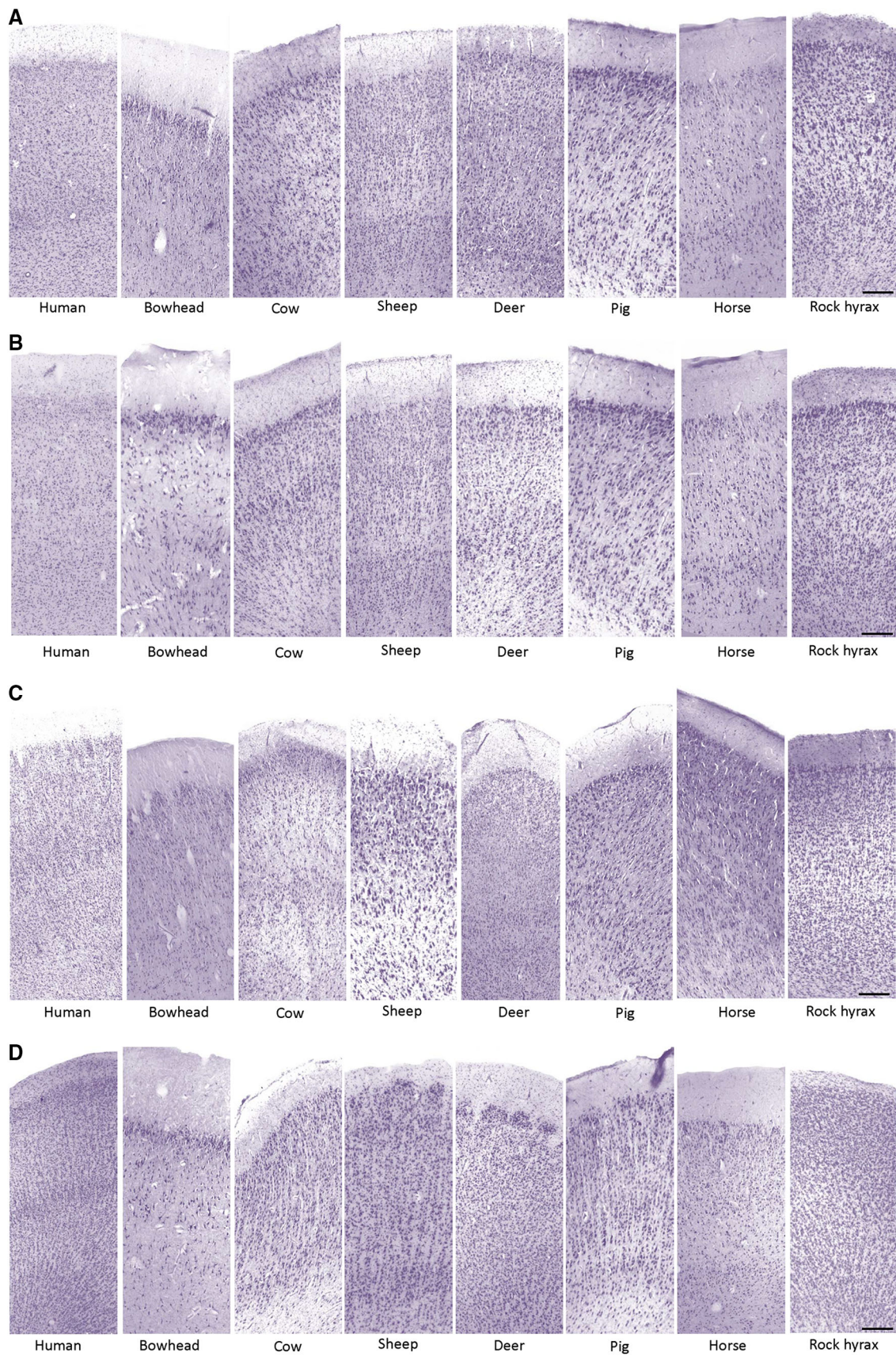


Fig. 2 Cytoarchitecture of (a) the frontal pole, (b) anterior cingulate cortex, (c) anterior insula, and (d) occipital pole for every species. Scale bars in the rightmost panel for each cortical region 250 μ m

Also notable was the sporadic occurrence of VENs and fork cells within layer III, but not layer V, of the rock hyrax (Fig. 5). With regard to distribution within each cortical area for all species, VENs were more numerous in the crowns of gyri and relatively sparse within sulcal depths consistent with the pattern of distribution reported for all other species with VENs (Nimchinsky et al. 1995; Cobos

and Seeley 2013; Butti and Hof 2010; Butti et al. 2009, 2011, 2013, 2014).

VENs and fork cells were found in layer V of all four cortical regions for the majority of the species examined here. The exceptions to this more ubiquitous distribution were the human (no VENs or fork cells in the frontal pole or occipital pole), rock hyrax (no VENs or fork cells in layer V and none within the occipital pole), and pig (no fork cells within the occipital cortex). Stereologic quantification of VENs and fork cells within layer V revealed species-specific densities and distributions. Figures 6 and 7

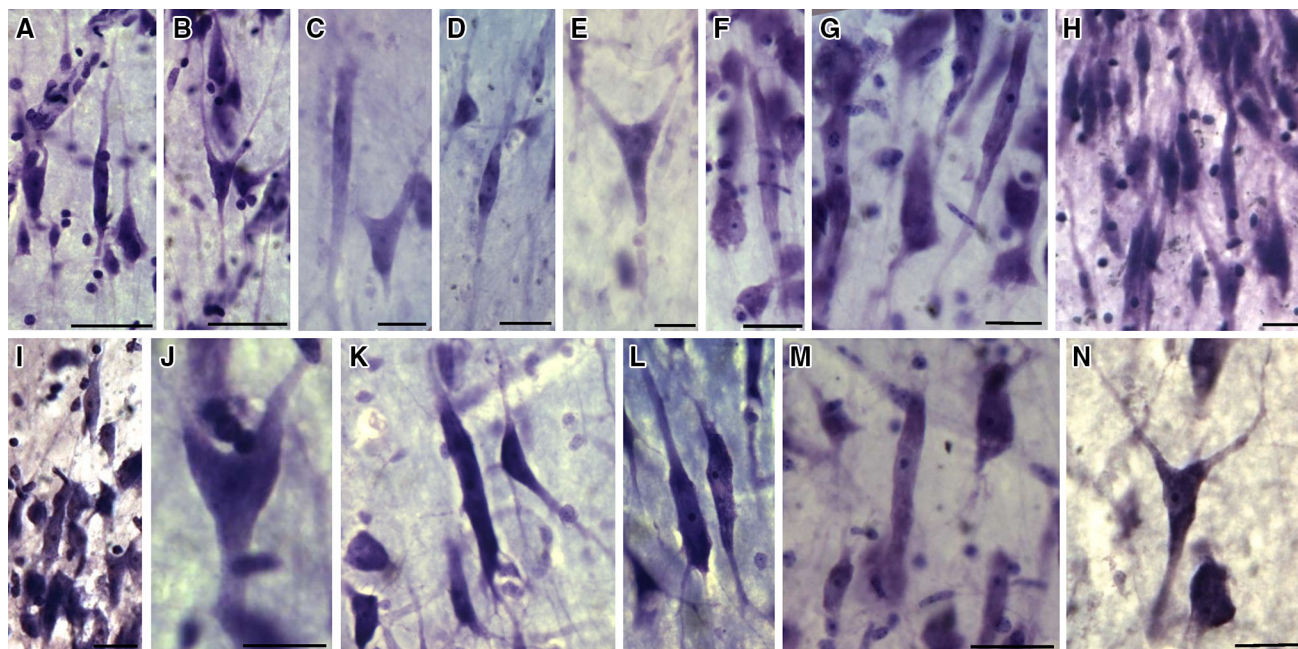


Fig. 3 Photomicrographs showing VENs and fork cells layer V of human (a, b), bowhead whale (c–e), calf (f, g), sheep (h), deer (i, j), pig (k, l), and horse (m, n). Scale bars 25 μ m

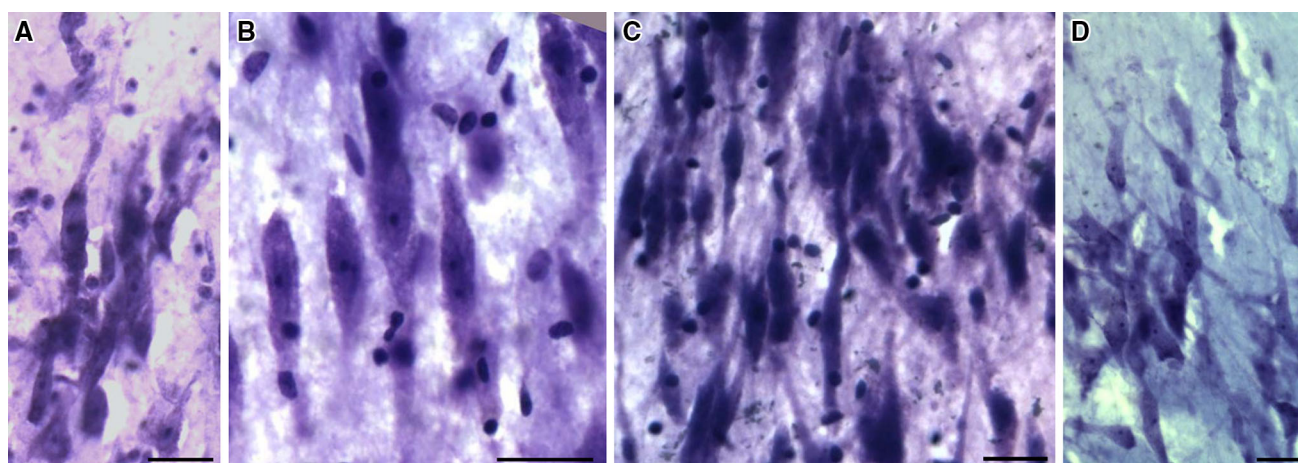


Fig. 4 Photomicrographs showing VENs in layer II of the calf (a), sheep (b, c), and bowhead whale (d). Notice the clustering of VENs in a, c, and d. Scale bars 25 μ m

Fig. 5 Photomicrographs showing a fork cell (a) and a VEN (b) in layer III of the frontal pole of the rock hyrax. Scale bars 25 μ m

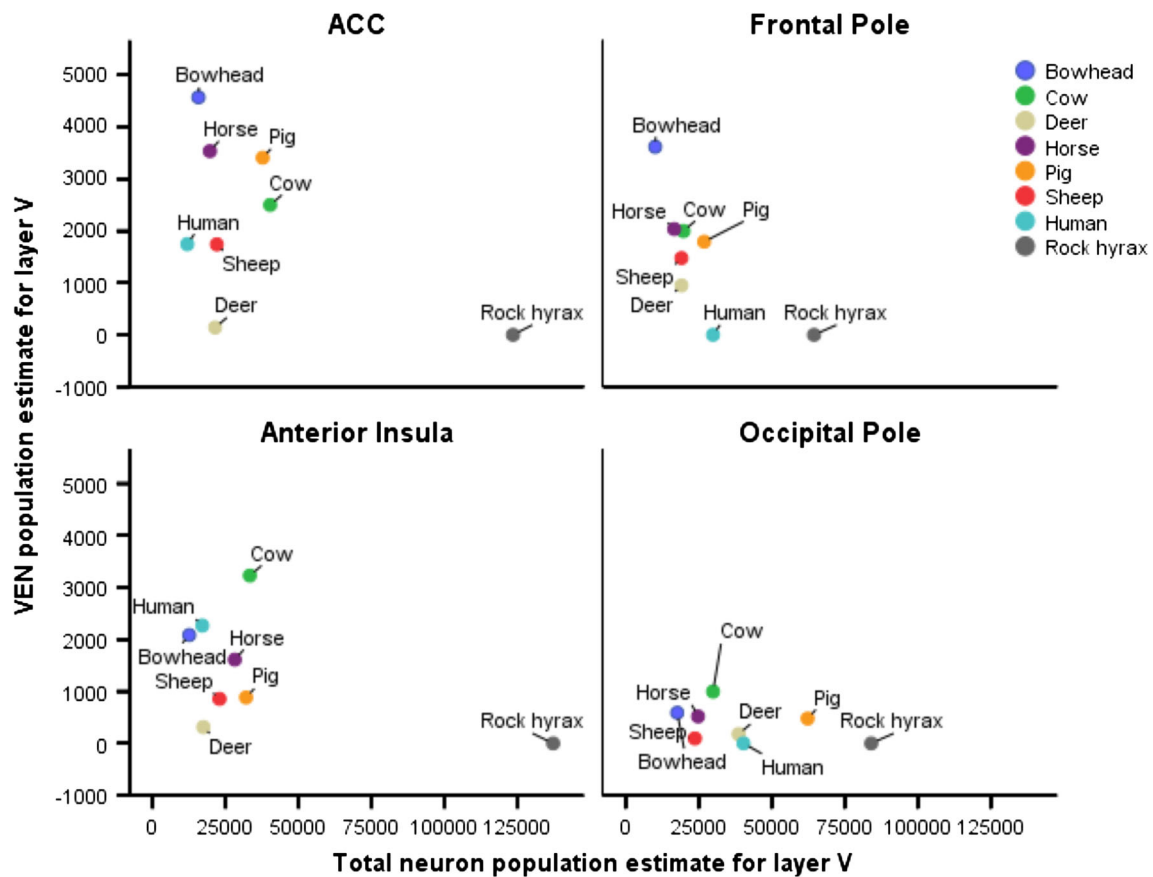
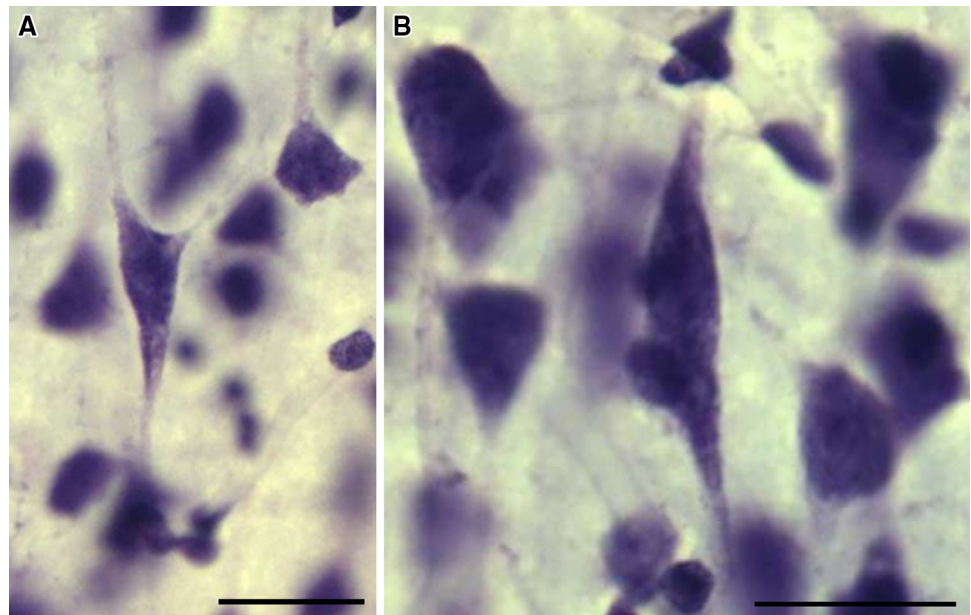


Fig. 6 Total VEN population estimates for layer V plotted as a function of total neuron population estimates of layer V for each cortical region. Note the high neuron population estimate for the rock hyrax with an absence of VENs

show the population estimates of VENs and fork cells within layer V plotted as a function of total neuron population estimates for each species and cortical region. There

was a significant negative correlation between each of these cell types and total neuron population within layer V of the frontal pole (VENs: Spearman's $\rho = -0.80$,

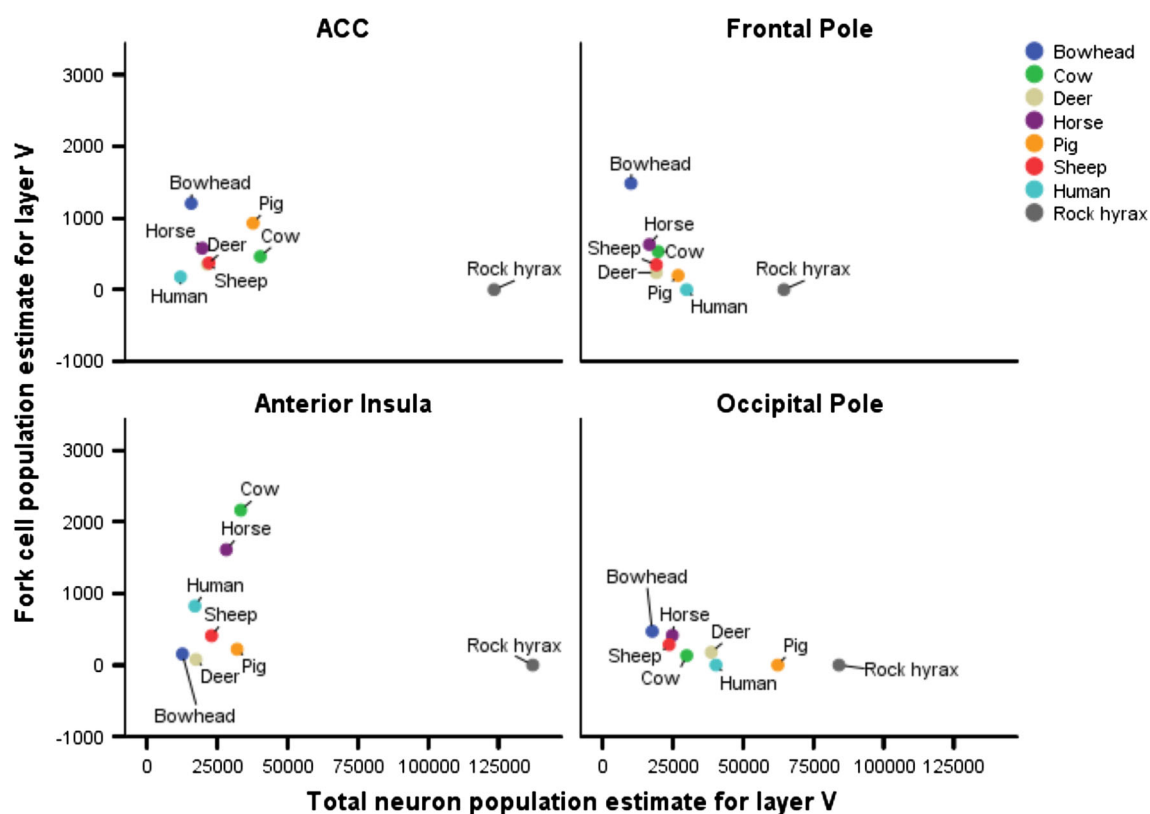


Fig. 7 Total fork cell population estimates for layer V plotted as a function of total neuron population estimates of layer V for each cortical region. Note the high neuron population estimate for the rock hyrax with an absence of fork cells

$p = 0.02$; fork cells: Spearman's $\rho = -0.92$, $p = 0.001$). There was also a significant negative correlation between fork cell and neuron number in the occipital lobe (Spearman's $\rho = -0.93$, $p = 0.001$). There were no other significant correlations between VENs and fork cells and total neuron estimates within layer V for the other cortical regions (all p 's > 0.05). The bowhead whale possessed the highest number of VENs and fork cells in layer V of the ACC and frontal pole, and the cow had the highest number of VENs and fork cells for the anterior insula. Within the occipital pole, the cow possessed the highest number of VENs, with the bowhead having a higher number of fork cells. When calculated as percentages of the total number of neurons within layer V, the bowhead whale possessed the highest percentage of VENs in all four cortical regions (Table 1). Further, the number of fork cells was positively correlated with the number of VENs in three of the regions (ACC: Spearman's $\rho = 0.91$, $p = 0.002$; frontal pole: Spearman's $\rho = 0.93$, $p = 0.001$; AI: Spearman's $\rho = 0.77$, $p = 0.02$); but not in the occipital pole (Spearman's $\rho = 0.54$, $p = 0.17$). These relationships are reflected in the scatter plots displayed in Fig. 8. The coincidence of VENs and fork cells is consistent with earlier reports of the association of these two cell types (Seeley et al. 2012; Ngowyang 1932, 1936).

Discussion

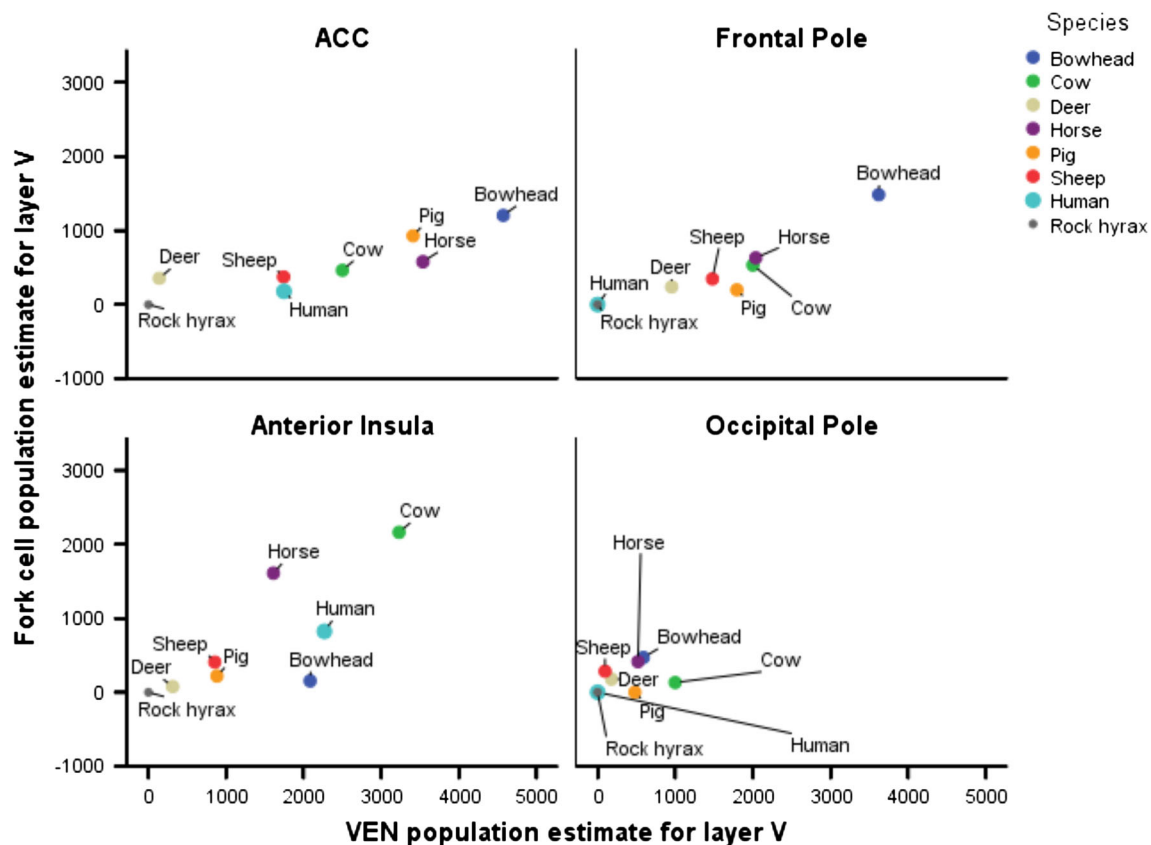
One goal of the present analysis was to evaluate a phylogenetically diverse range of species for the presence of VENs and fork cells to gain a better perspective on the evolution of these neuron types. VENs and fork cells were present in each of the species examined here, with species-specific distributions and densities within the four cortical regions chosen for study. Previous comparisons of VEN densities that included humans, great apes, cetaceans, and elephants suggested that VENs were most numerous in human ACC and FI (Nimchinsky et al. 1999; Hakeem et al. 2009; Butti et al. 2009). However, in the present comparisons, the human did not possess the highest density of either VENs or fork cells in those cortical regions where they were found (ACC and AI). Further, while VENs and fork cells were selectively distributed in humans, these cells were more ubiquitous in many of the species examined here, a finding consistent with recent data from the pygmy hippopotamus (Butti et al. 2011, 2014).

Among cetaceans, VENs have been observed in mysticetes, including the humpback whale (*Megaptera novaeangliae*) and fin whale (*Balaenoptera physalus*) (Hof and Van der Gucht 2007), as well as in odontocetes, including the sperm whale (*Physeter macrocephalus*), killer whale

Table 1 Percentage of VENs and fork cells in layer V of each cortical region

Species	Frontal pole		ACC		Anterior insula		Occipital pole	
	% VEN	% Fork cells	% VEN	% Fork cells	% VEN	% Fork cells	% VEN	% Fork cells
Bowhead whale	24	10	21	6	14	1	3	3
Rock hyrax	0	0	0	0	0	0	0	0
Cow	9	2	6	1	8	6	3	0.4
Pig	6	1	8	2	3	1	1	0
Deer	5	1	5	1	2	0.4	0.5	0.5
Sheep	7	2	7	2	4	2	0.4	1
Horse	11	3	15	2	5	5	2	2
Human	0	0	13	1	11	4	0	0

Percentages were calculated as the population estimate of VENs or fork cells divided by the total neuron population estimates

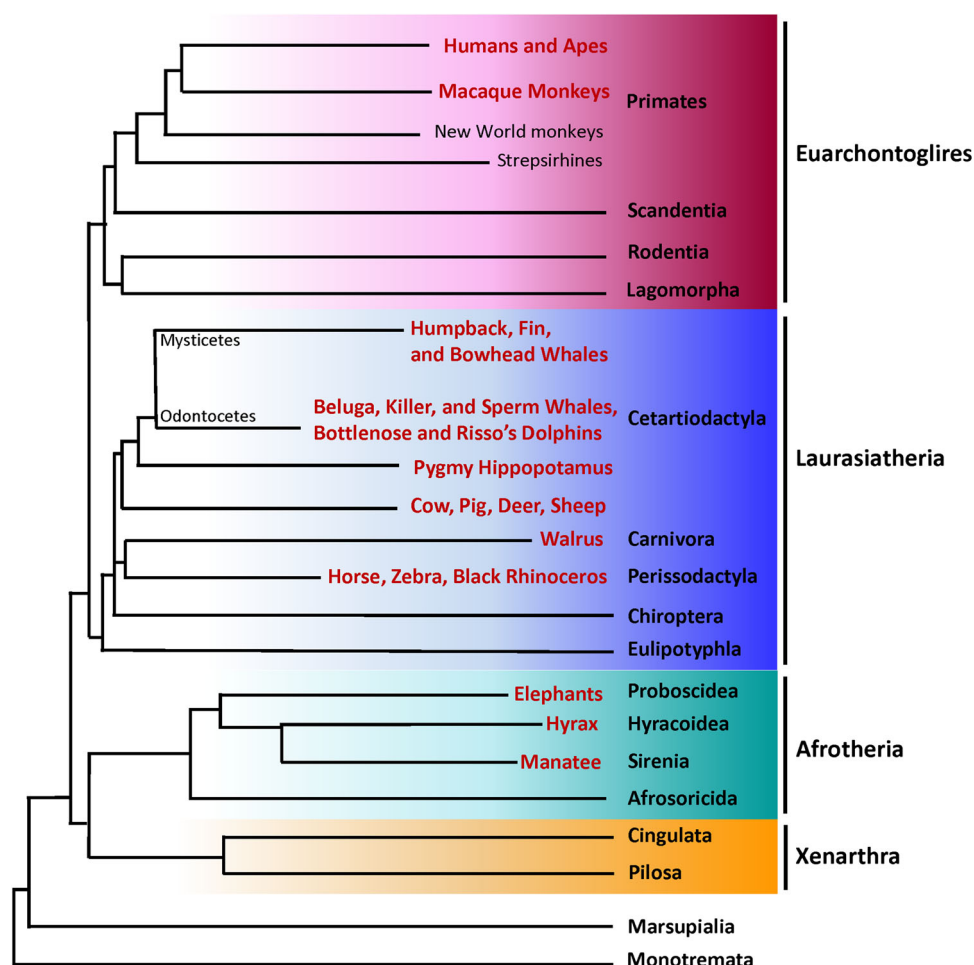
**Fig. 8** Fork cell population estimates plotted as a function of the VEN population estimates. Note the positive relationship between each population for every cortical region

(*Orcinus orca*), bottlenose dolphin (*Tursiops truncatus*), Risso's dolphin (*Grampus griseus*), and the beluga whale (*Delphinapterus leucas*) (Hakeem et al. 2009; Butti et al. 2009). For each of these cetacean species, there was a regionally limited cortical distribution of VENs that included the ACC, anterior insula, and frontopolar cortex. Our finding of VENs in every cortical region of the bowhead whale deviates from this pattern. Odontocetes and mysticetes are monophyletic (Spaulding et al. 2009; Gatesy and O'Leary 2001) and derived from artiodactyls

(Spaulding et al. 2009; Hans Thewissen et al. 2007). Bowhead whales are clearly mysticetes and are not basal to any odontocete (Nikaido et al. 2006; Sasaki et al. 2005). They are the least encephalized cetaceans (Hans Thewissen et al. 2011), and may be in a basal position among the mysticetes (Marx 2011; Price et al. 2005).

In recognition of their close phylogenetic relationship, cetaceans and artiodactyls (even-toed ungulates) are grouped together into the clade Cetartiodactyla (Ruben et al. 2012). The discovery of VENs in every neocortical region of

Fig. 9 Phylogeny of living mammals (adapted from Song et al. 2012) showing the phylogenetic distribution of VENs. The common names for those species in which VENs have been observed are shown in red



the pygmy hippopotamus, one of the closest terrestrial relatives of cetaceans based on molecular phylogenetics, suggested that VENs might be present in a wider range of species than previously thought. Our sample also included the rock hyrax, the closest living relative of elephants, as well as a perissodactyl (the horse) because VENs were previously observed in other perissodactyls, including the common zebra (*Equus burchellii*) and black rhinoceros (*Diceros bicornis*), in addition to a carnivore, the walrus (*Odobenus rosmarus*) (Butti and Hof 2010; Butti et al. 2011; Raghanti et al. 2014). A graphic representation of the phylogenetic distribution of VENs is provided in Fig. 9.

Interestingly, the calf was the only juvenile individual in this sample and possessed a high density of VENs and fork cells, particularly within the AI (see Figs. 6, 7, 8). The majority of VENs emerge during postnatal life in humans, indicating that the maturation of this neuron population is delayed during human development. In contrast, a high number of VENs is already present in the fetal chimpanzee (Allman et al. 2010; Hayashi et al. 2001). Additional ontogenetic analyses that include a wider range of species are needed to determine whether a delayed maturation of VENs is exclusive to humans or if this is characteristic of this neuron type.

Many studies have attempted to determine the phenotype and potential function of VENs, although most comparative research has been limited to primate species (Stimpson et al. 2011; Cobos and Seeley 2013; Allman et al. 2010). In terms of biochemical phenotype, human VENs are immunoreactive for activating transcription factor 3 (ATF3); a protein involved in pain sensitivity, interleukin-4 receptor alpha chain (IL4R α) that mediates allergic reactions; and neuromedin B that is also expressed in the gut and was suggested as a potential contributor in connecting visceral states with social awareness (Allman et al. 2010). A comparative analysis that included humans, great apes, and lesser apes found that a greater percentage of human VENs expressed IL4R α and ATF3, potentially contributing to differences in interoceptive sensitivity that contributes to social intelligence (Stimpson et al. 2011). Future studies that extend sampling to include artiodactyls are required to clarify these phenotypic differences among species in terms of functional significance.

The presence of VENs and fork cells in such a diverse range of species suggests that these neuron morphologies represent specializations of pyramidal neurons that readily emerge within individual clades or even individual species

in response to some functional requirement. Earlier evidence that these are subcortically projecting neurons (Nimchinsky et al. 1995; Allman et al. 2010; Evrard et al. 2012) was further supported by the recent finding that VENs preferentially express transcription factors that are characteristic of subcortical projection neurons rather than callosal or corticothalamic projections (Cobos and Seeley 2013). The observation of higher densities of VENs at the crown of gyri and sparse distribution within sulcal depths among all phylogenetic groups indicates that their presence may be related to the degree of convolution of the cortical surface. Pillay and Manger (2007) compared amount of gyrification for different terrestrial mammalian orders and found that ungulates have the greatest gyrification as well as the thinnest cerebral cortex. Cetaceans are highly gyrencephalic and possess a thin cortex (Welker 1990; Manger et al. 2012; Ridgway and Brownson 1984). Coincident with the increased gyrification and thin cortex is a low neuronal density for both artiodactyls and cetaceans (Manger 2006; Chow et al. 1950). The preferential location of VENs at the crowns of gyri may be related to the mechanical stressors associated with convolution, as detailed by Van Essen's theory of tension-based morphogenesis of the brain (Van Essen 1997). Van Essen speculated that the neurons of the deep layers within crown gyri would be stretched radially, and it is possible that pyramidal neurons in gyrencephalic brains are unable to resist this tension and are stretched into the VEN morphotype. A gyrencephalic brain is a common feature among species that possess high densities of VENs. The lissencephalic manatee (Reep and O'Shea 1990) does possess VENs, but as in our observations in the similarly lissencephalic rock hyrax, their distribution is very sparse and limited in cortical distribution and restricted to layer III (Butti et al. 2009). Also noteworthy in the rock hyrax was the extremely high neuron density relative to the other species in our sample (see Figs. 6, 7). It is possible that cortical thickness, gyrification, and neuron densities are influential factors in determining presence or absence of VENs and fork cells within the cerebral cortex, and additional research will be necessary to explore those prospects.

There are several explanations possible to account for the phylogenetic distribution of VENs and fork cells. One possibility is that they are an ancient neuron type that evolved early in mammals and was subsequently lost in many species. An alternative is convergent evolution of VENs and fork cells in Euarchontoglires, Laurasiatheria, and Afrotheria. The most parsimonious explanation in this model would be that VENs and fork cells emerged late in both euarchontoglires and afrotherians, as they are present in only a few species and with a distribution restricted to a few cortical areas (see Fig. 9). In laurasiatherians, VENs and fork cells emerged early, but the occurrence of

differential distribution characteristics among the taxonomic orders is not clearly related to phylogeny. It is possible that an ubiquitous cortical distribution is plesiomorphic for laurasiatherians, and the restricted distribution observed in the odontocetes and most mysticetes is apomorphic. Future research that includes additional species is needed to clarify these relationships. Nevertheless, the repeated evolution of VENs and associated fork cells suggests that these neuronal subtypes represent a relatively simple morphological response to functional requirements that may be related to the degree of cortical gyrification. The presence of VENs and fork cells in the species analyzed here affords the opportunity for additional empirical study of their structural and functional relationships.

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